Train, Validation, Test

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One readable role-and-group evidence record.

Explain selection,assign actions and audit source-separated counts.

Keep one habit: W1–W3/W5 → selected P/Q/Rtask → project split record → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Absolute Error and MAE. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.

Hold Out Test Data

> **My remembered idea, example and correction**

Absolute Error and MAE

> **My remembered idea, example and correction**

Error Analysis

> **My remembered idea, example and correction**

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main candidates training MAE A1/B0/C2 cm, validation A3/B5/C3 cm. Rule:smallest validation MAE, then simpler; Adeclared simpler than C, so choose A. Picture A/Btraining1/0 cm, validation 2/3 cm; choose A. Pictured validation 2/2 tie still chooses Aunder predeclared simplicity. Group manifestL01–L18 train54 images, L19–L24 validation18, L25–L30 test18; 30 leaves×3 views=90 images.

  Fit learned transforms on training; use validation for recorded choices; freeze before the final check. No final score is supplied.

- Reference for this exercise: open this week's student lesson, "Train, Validation, Test", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Apply the worked **selection rule**. Explain why B is not chosen.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main candidates training MAE A1/B0/C2 cm, validation A3/B5/C3 cm. Rule:smallest validation MAE, then simpler; Adeclared simpler than C, so choose A. Picture A/Btraining1/0 cm, validation 2/3 cm; choose A. Pictured validation 2/2 tie still chooses Aunder predeclared simplicity. Group manifestL01–L18 train54 images, L19–L24 validation18, L25–L30 test18; 30 leaves×3 views=90 images.

  Fit learned transforms on training; use validation for recorded choices; freeze before the final check. No final score is supplied.

- Where should parameter **fitting**, setting selection and final reporting occur?

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main candidates training MAE A1/B0/C2 cm, validation A3/B5/C3 cm. Rule:smallest validation MAE, then simpler; Adeclared simpler than C, so choose A. Picture A/Btraining1/0 cm, validation 2/3 cm; choose A. Pictured validation 2/2 tie still chooses Aunder predeclared simplicity. Group manifestL01–L18 train54 images, L19–L24 validation18, L25–L30 test18; 30 leaves×3 views=90 images.

  Fit learned transforms on training; use validation for recorded choices; freeze before the final check. No final score is supplied.

- A team creates D after reading **validation** mistakes. What changes in the record?

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main candidates training MAE A1/B0/C2 cm, validation A3/B5/C3 cm. Rule:smallest validation MAE, then simpler; Adeclared simpler than C, so choose A. Picture A/Btraining1/0 cm, validation 2/3 cm; choose A. Pictured validation 2/2 tie still chooses Aunder predeclared simplicity. Group manifestL01–L18 train54 images, L19–L24 validation18, L25–L30 test18; 30 leaves×3 views=90 images.

  Fit learned transforms on training; use validation for recorded choices; freeze before the final check. No final score is supplied.

- Why is the winning **validation** score not a guaranteed future score?

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- This selected task uses fresh P/Q/Rtraining and validation MAEs, with Pdeclared simpler than R. Apply the stated validation-first rule and tie-break; do not substitute the main A/B/Cnumbers. No final-test values are supplied, and none are needed to select the candidate or state the freeze-and-test action.

- **Training** errors P 2 Q 0 R 1; **validation** errors P 4 Q 6 R 4. Rule: smallest **validation** MAE, then simpler; P is simpler than R. Choose and state the test action.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply independently

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Publish the grouped **split strategy** and counts for the Plant Classifier, including why plant identity stays within one split.

- The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Assign data roles for **Validation** and model selection. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for **Validation** and model selection. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- You have 30 leaves with 3 views each. Allocate 18 leaves to train,6 to **validation**,6 to test. How many images are in each? Where is a learned normalisation mean fitted?

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Split 100 independent leaves into 60 train,20 **validation**,20 test.
- Explain roles and pre**process**ing.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Assign roles: set A fits model parameters; B selects between two models; C is opened once after choices are fixed. Which is train, **validation** and test?
- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 24 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- You have 30 leaves with 3 views each. Allocate 18 leaves to train,6 to **validation**,6 to test. How many images are in each? Where is a learned normalisation mean fitted?

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- **Training** errors P 2 Q 0 R 1; **validation** errors P 4 Q 6 R 4. Rule: smallest **validation** MAE, then simpler; P is simpler than R. Choose and state the test action.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Assign a separate job to **training**, **validation** and final test **evidence**.
- Candidates A and B have **training** MAE1 and 0 cm; **validation** MAE2 and 3 cm. **Selection rule** is smallest **validation** MAE.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Why does B's zero **training** MAE not settle the original choice? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/24

Train, Validation, Test

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Fit on training

Both candidates are built using training data; B fits it more closely.



2. Select on validation

Choose A because2 cm is below3 cm.



3. Reserve final test

Freeze A before scoring final cases; do not select again using their labels.

CHANGE ONE THING

A and B both have validation MAE2 cm; A is simpler under a predeclared tie rule.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Assign a separate job to **training**, **validation** and final test **evidence**.
- Which labelled step matches this detail? Freeze A before scoring final cases; do not select again using their labels.
- 1. Reserve final test
- 2. Fit on **training**
- 3. Select on **validation**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Reserve final test Freeze A before scoring final cases; do not select again using their labels. Read the picture in order.

Freeze A before scoring final cases; do not select again using their labels. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: A and B both have **validation** MAE2 cm; A is simpler under a predeclared tie rule. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. Select A using the predeclared simpler-model tie-break.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Select A using the predeclared simpler-model tie-break. The **validation** errors tie, so the stated secondary rule decides without consulting final data.

Repair and keep the record

Review Hold Out Test Data (ai-4/23). Goal: Freeze the full evaluation procedure before testing. Short practice: What is needed for an honest final claim about the revised version?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Absolute Error and MAE (ai-4/21). Goal: Calculate MAE MSE and RMSE. Short practice: Why is quoting MSE as4.67 cm misleading?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Error Analysis (ai-4/18). Goal: Categorize development errors using **evidence**. Short practice: What **evidence** would support the claim that a change helps fresh dim-light images?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

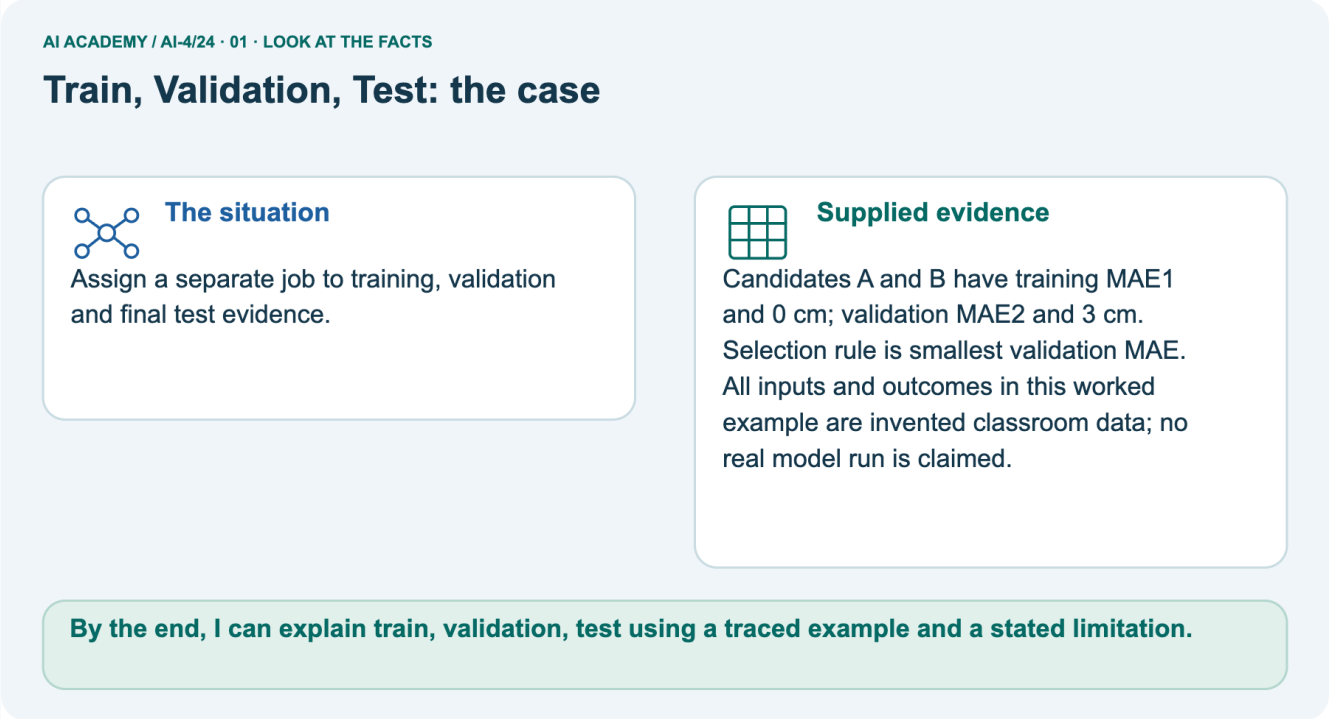
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. Train, Validation, Test: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

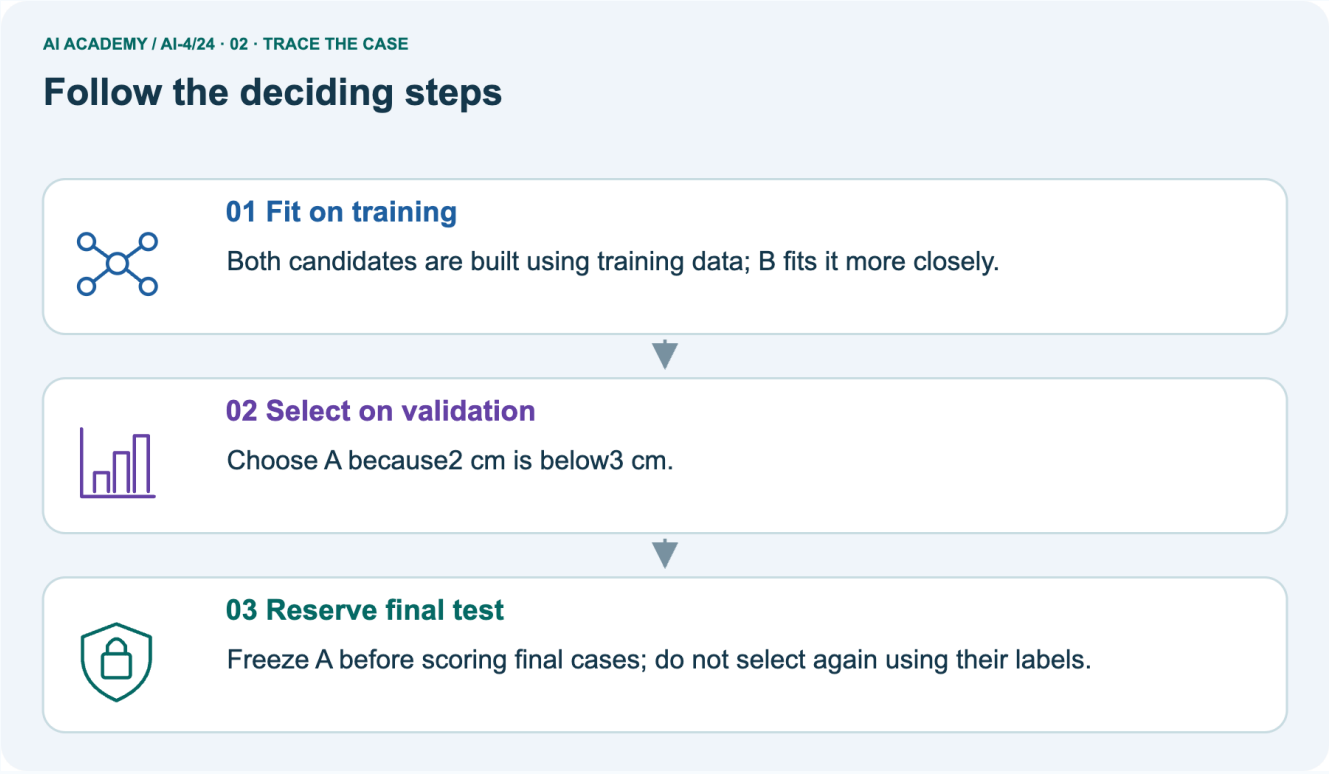
Assign a separate job to **training**, **validation** and final test **evidence**. Supplied facts: Candidates A and B have **training** MAE1 and 0 cm; **validation** MAE2 and 3 cm. **Selection rule** is smallest **validation** MAE.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Fit on **training**: Both candidates are built using **training** data; B fits it more closely. → Select on **validation**: Choose A because2 cm is below3 cm. → Reserve final test: Freeze A before scoring final cases; do not select again using their labels.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Validation and model selection at a glance

AI ACADEMY / AI-4/24 · 03 · INSPECT THE LESSON MODEL

Validation and model selection at a glance

Part	Evidence or action	Interpretation
Training	Fit each candidate	No test labels
Validation	Choose A by rule	Record every attempt
Final test	Frozen model only	Report after selection

A tie policy prevents a convenient choice after inspecting results. Equal validation scores do not prove equal future performance. Record candidate settings, split identifiers, preprocessing and the full comparison, including rejected options.

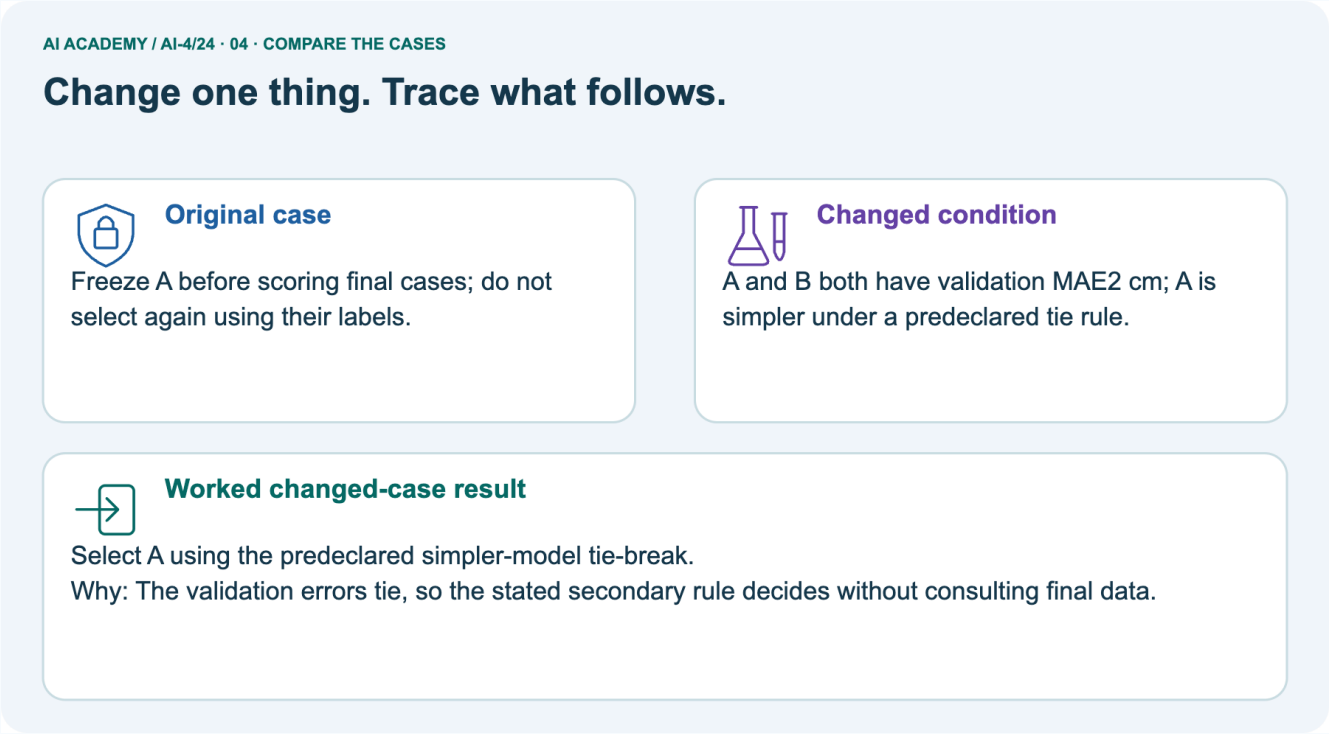
Follow the labelled picture with the matching explanation. Open the vector diagram

Validation and model selection at a glance. **Training** | Fit each candidate | No test labels. **Validation** | Choose A by rule | Record every attempt. Final test | Frozen model only | Report after selection

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

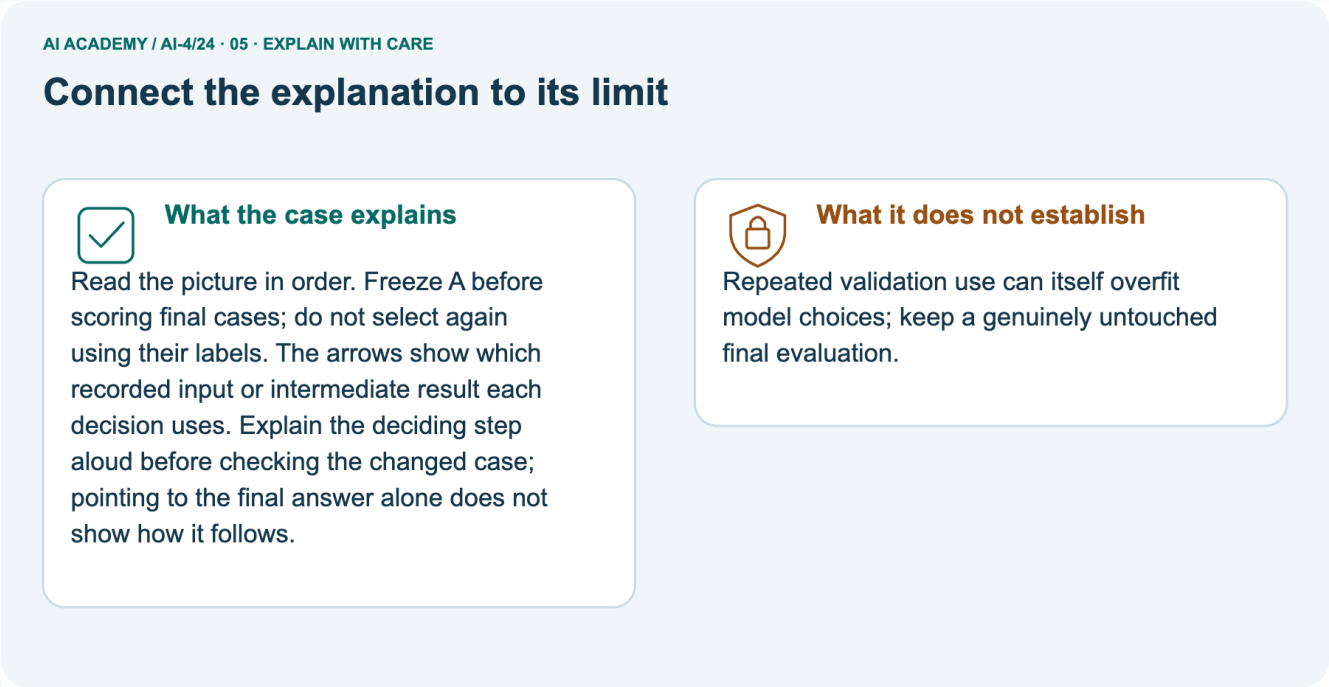
Original case: Freeze A before scoring final cases; do not select again using their labels. Changed condition: A and B both have **validation** MAE2 cm; A is simpler under a predeclared tie rule.

Expected result: Select A using the predeclared simpler-model tie-break. Reason: The **validation** errors tie, so the stated secondary rule decides without consulting final data.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Freeze A before scoring final cases; do not select again using their labels. The arrows show which recorded **input** or intermediate result each decision uses.

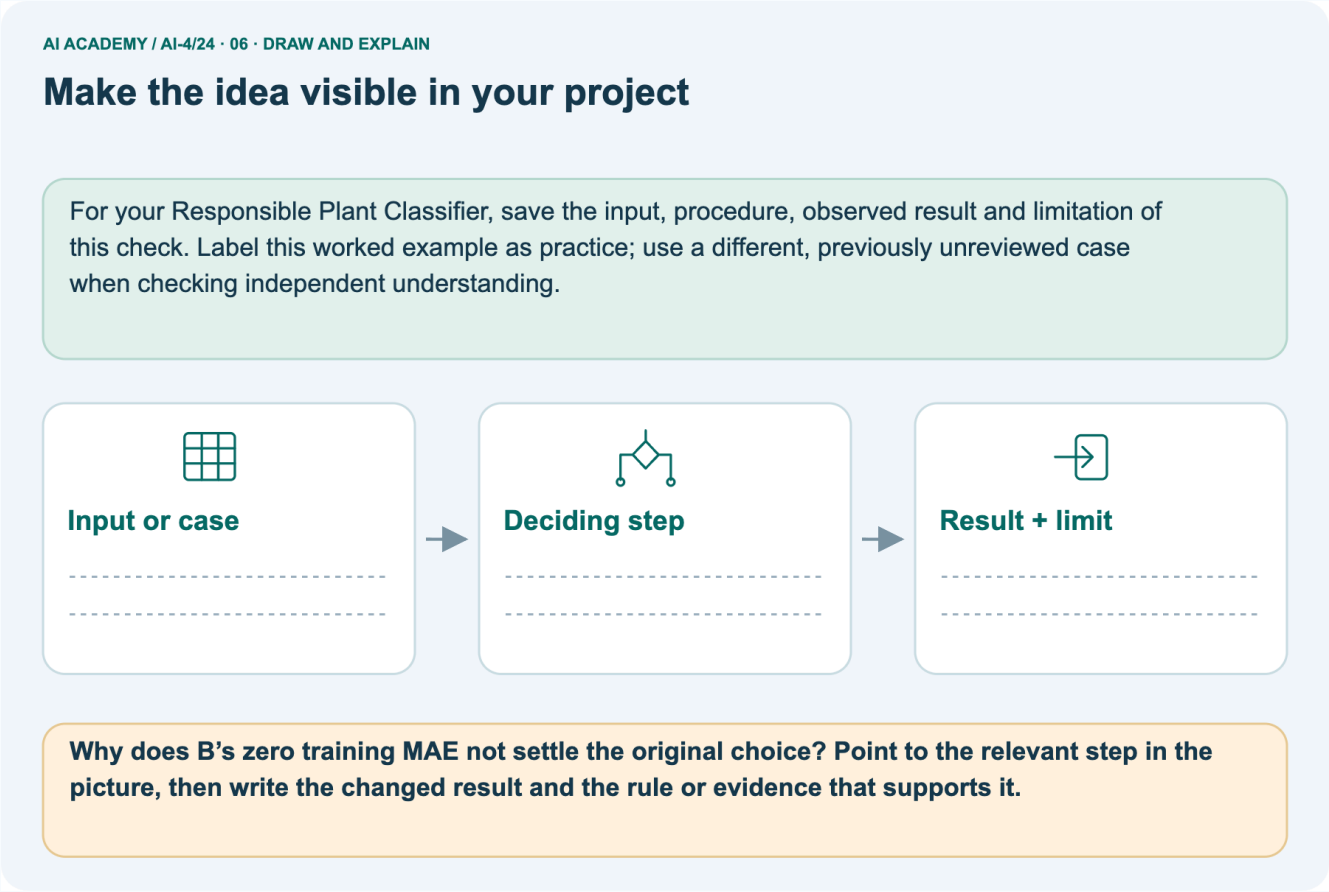
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Repeated **validation** use can itself overfit model choices; keep a genuinely untouched final evaluation.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why does B's zero **training** MAE not settle the original choice? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

01. Train, Validation, Test: the case

02. Follow the deciding steps

03. Validation and model selection at a glance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project
</details>

<details>
<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.
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